Battle of the generations

***p. 160* R. L. Trivers, in 1972, neatly solved the problem …**

Robert Trivers, whose papers of the early 1970s were among the most important inspirations for me in writing the first edition of this book, and whose ideas especially dominate Chapter 8, has finally produced his own book, *Social Evolution*. I recommend it, not only for its content but for its style: clear-thinking, academically correct but with just enough anthropomorphic irresponsibility to tease the pompous, and spiced with personal autobiographical asides. I cannot resist quoting one of these: it is so characteristic. Trivers is describing his excitement on observing the relationship between two rival male baboons in Kenya: 'There was another reason for my excitement and this was an unconscious identification with Arthur. Arthur was a superb young male in his prime … ' Trivers's new chapter on parent–offspring conflict brings the subject up to date. There is indeed rather little to add to his paper of 1974, apart from some new factual examples. The theory has stood the test of time. More detailed mathematical and genetic models have confirmed that Trivers's largely verbal arguments do indeed follow from currently accepted Darwinian theory.

***p. 175* According to him the parent will always win.**

Alexander has generously conceded, in his 1980 book *Darwinism and Human Affairs* (p. 39), that he was wrong to argue that parental victory in parent–offspring conflict follows inevitably from fundamental Darwinian assumptions. It now seems to me that his thesis, that parents enjoy an asymmetrical advantage over their offspring in the battle of the generations, could be bolstered by a different kind of argument, which I learnt from Eric Charnov.

Charnov was writing about social insects and the origins of sterile castes, but his argument applies more generally and I shall put it in general terms. Consider a young female of a monogamous species, not necessarily an insect, on the threshold of adult life. Her dilemma is whether to leave and try to reproduce on her own, or to stay in the parental nest and help rear her younger brothers and sisters. Because of the breeding habits of her species, she can be confident that her mother will go on giving her full brothers and sisters for a long time to come. By Hamilton's logic, these sibs are just as genetically 'valuable' to her as her own offspring would be. As far as genetic relatedness is concerned, the young female will be indifferent between the two courses of action; she doesn't 'care' whether she goes or stays. But her parents will be far from indifferent to which she does. Looked at from the point of view of her old mother, the choice is between grandchildren or children. New children are twice as valuable, genetically speaking, as new grandchildren. If we speak of conflict between parents and offspring over whether the offspring leaves or stays and helps at the nest, Charnov's point is that the conflict is a walk-over for the parents for the very good reason that only the parents see it as a conflict at all!

It is a bit like a race between two athletes, where one has been offered £1,000 only if he wins, while his opponent has been promised £1,000 whether he wins or loses. We should expect that the first runner will try harder and that, if the two are otherwise evenly matched, he will probably win. Actually Charnov's point is stronger than this analogy suggests, because the costs of running flat out are not so great as to deter many people, whether they are financially rewarded or not. Such Olympic ideals are too much of a luxury for the Darwinian games: effort in one direction is always paid for as lost effort in another direction. It is as if the more effort you put into any one race, the less likely you are to win future races because of exhaustion.

Conditions will vary from species to species, so we can't always forecast the results of Darwinian games. Nevertheless, if we consider only closeness of genetic relatedness and assume a monogamous mating system (so that the daughter can be sure that her sibs are full sibs), we can expect an old mother to succeed in manipulating her young adult daughter into staying and